

cann you show me an example, why poisson, french fore fish

1. Etymology: Why the name "Poisson"?

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2. Why use a Poisson model instead of standard linear regression?

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PNG

Non-negative integers only: Real event counts are $Y \in \{0, 1, 2, 3, \dots\}$. A standard linear regression line can output impossible values like -1.4 intrusions or 2.7 vehicles.

PNG+ 1

Variance grows with the mean: In count data, larger average rates typically come with wider spread ($\text{Var}(Y) = E(Y) = \lambda$). Linear regression assumes constant variance (homoscedasticity), which breaks down on rate data.

PNG

Rare event modeling: It accurately calculates the probability of rare anomalies when you know the baseline arrival rate.

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PNG

$$\Pr(Y=k)=$$

$$k!$$

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$$-\lambda$$

$$\lambda$$

$$k$$

for $k=0,1,2,\dots$

PNG

3. Concrete Example: Perimeter Crossing Telemetry

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0 events (Quiet perimeter):

$$\Pr(Y=0)=$$

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$\approx 0.135 (13.5\%)$

1 event:

$\Pr(Y=1)=$

$1!$

e

-2

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$$=2e$$

$$-2$$

$$\approx 0.271 (27.1\%)$$

2 events (Expected average):

$$\Pr(Y=2)=$$

$$2!$$

$$e$$

$$-2$$

$$\cdot 2$$

$$2$$

$$=2e$$

$$-2$$

$$\approx 0.271 (27.1\%)$$

3 events:

$$\Pr(Y=3)=$$

$$3!$$

$$e$$

$$-2$$

$$\cdot 2$$

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$$e$$

$\approx 0.180(18.0\%)$

The Anomaly Trigger (Why this saves bandwidth)

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$$\Pr(Y=7)=$$

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$$\cdot 2$$

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$$128 \cdot e$$

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$\approx 0.0034(0.34\%)$

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Normal windows ($k \leq 4$): Evaluated as benign ambient noise; suppressed completely from the cellular/radio uplink.

Anomaly window ($k=7$): Flagged instantly at the edge as a statistically significant departure from $\lambda=2$, prioritizing a compact state vector burst to the command center without having streamed continuous video.

an attack by a group of hamsters

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The Tactical Setup

Sensor Sector: Perimeter North, Zone 4 (Chain-link breach zone).

Sensor Baseline: Under normal conditions at 03:00 AM, wind-blown tumbleweeds and nocturnal rodents produce a baseline arrival rate of:

$\lambda=0.5$ kinetic crossings per 5-minute window

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PNG

$$\Pr(Y=k)=$$

$$k!$$

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PNG

Nominal Activity Probabilities ($\lambda=0.5$)

0 crossings (Nominal quiet):

$$\Pr(Y=0)=$$

$$0!$$

e

$$-0.5$$

$$\cdot (0.5)$$

$$0$$

$$=e$$

$$-0.5$$

$$\approx 0.6065 (60.7\%)$$

1 crossing (Lone foraging rodent):

$$\Pr(Y=1)=$$

$$1!$$

$$e$$

$$-0.5$$

$$\cdot (0.5)$$

$$1$$

$$=0.5 \cdot e$$

$$-0.5$$

$$\approx 0.3033 (30.3\%)$$

2 crossings (Two independent critters):

$$\Pr(Y=2)=$$

$$2!$$

e

-0.5

$\cdot (0.5)$

2

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0.25

$\cdot e$

-0.5

$\approx 0.0758 (7.6\%)$

Nominal Threshold: $\Pr(Y \leq 2) = 60.7\% + 30.3\% + 7.6\% = 98.6\%$.

Edge Action: Evaluated as ambient environmental noise. Local edge NPU suppresses all uplink transmissions. 0 bytes sent over satellite/cellular.

The Anomaly: Coordinated Hamster Incursion ($k=8$)

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$$\Pr(Y=8)=$$

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$$\begin{aligned} & (0.5)^8 \\ & = 0.00390625 \\ & 8! = 40,320 \\ & e^{-0.5} \\ & \approx 0.60653 \end{aligned}$$

$$\Pr(Y=8)=$$

$$\frac{0.00390625}{40,320}$$

$$0.60653 \cdot 0.00390625$$

$$\approx$$

$$\frac{0.002369}{40,320}$$

$$0.002369$$

$$\approx 5.88 \times 10^{-6}$$

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Edge Decision Matrix Execution

Anomaly Validation: $p=0.0000000588 \ll \alpha$

threshold

(0.001). The Poisson filter proves this is not ambient fauna drift.

Kinematic Feature Discrimination: The state vector indicates:

Average bounding box area: 0.015 m

2

(Rodent class, non-human).

Velocity: 0.8 m/s forward swarm trajectory.

Escalation Protocol:

Asymmetric cost matrix confirms non-hostile mass (no kinetic countermeasure deployed).

Telemetry payload: 48-byte Protobuf packet pushed to command dashboard:

```
{sector: 4, event: SWARM_ANOMALY, count: 8,  
confidence: 0.9999, class: MICRO_FAUNA}
```

Egress used: **48 bytes** instead of streaming a 200 MB 4K video file of hamsters running across the tarmac.

this would make a funny animation for the pitch

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Frame 1: The Status Quo (Cloud-First Video)

Visual: A split-screen showing a 4K camera feed of a gravel yard at night. Suddenly, eight tiny hamsters in tactical formation scurry across the frame.

On-Screen Metrics:

Uplink Stream: 25 Mbps H.265 saturated.

Cloud Pipeline: Ingesting 200 MB of video.

Latency: Buffering... 4.2 seconds.

Cost: Cellular roaming charges ticking up.

Callout: "The cloud spends 200 MB streaming high-res gravel and rodent fur to tell you something moved."

Frame 2: The VectorEdge Approach (Edge-Native Telemetry)

Visual: The camera feed is dark/muted, overlaid with minimal wireframe vector bounding boxes tracking the cluster.

The Math Overlay:

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baseline

$=0.5 \Rightarrow k=8 \Rightarrow p=5.88 \times 10$

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On-Screen Metrics:

Edge Classification: $< 8 \text{ ms}$

Egress Payload: 48 bytes (Protobuf telemetry packet)

Classification: MICRO_FAUNA_SWARM (Suppressed /
Non-Kinetic)

Callout: "VectorEdge evaluates the statistical anomaly in 8 milliseconds on a \$15 edge chip and sends 48 bytes."

Why It Works for YC & Defense Investors

Disarms the room with wit: It replaces dry, generic surveillance footage with a memorable scenario everyone immediately understands.

Proves the math instantly: Showing the Poisson anomaly formula on screen proves you aren't just selling "AI buzzwords"—you have a rigorous statistical foundation built for low-SWaP compute.

Drives home the bandwidth disparity: Contrasting **200 Megabytes vs. 48 Bytes** is the exact quantitative punchline that sticks with technical partners and partners at In-Q-Tel or Y Combinator.

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then have a cat drops by a quad copter in a tactical vest at yard endge

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Frame 3: Airborne Ingress & Tactical Deployment (T + 4.0s)

Visual: A small quadcopter hovers into the upper periphery over Sector 4. A tactical cat in a miniature plate carrier fast-ropes / drops down into the gravel yard edge to intercept the hamster swarm.

The Cloud-AI Video Reaction:

Status: Cloud decoder chokes on the rapid frame delta (spinning loading wheel).

Alert Text: MOTION DETECTED (Unknown Object - 84% confidence)

Bandwidth Spike: Egress surges to 12 Mbps trying to encode high-speed rotor blur and gravel dust.

Operator Interface: Alert delayed by 3.8 seconds. Operator still staring at stale pixel frames.

Frame 4: VectorEdge Edge-Native Kinematic Decomposition (T + 4.01s)

Visual: The camera feed is rendered as a clean, low-poly wireframe vector field. The edge runtime instantly tracks two distinct kinematics profiles simultaneously without touching the cloud.

Vector Deconstruction (On-Screen HUD):

Primary Vector (Airborne UAV Platform):

Velocity: (v

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, v

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$\mathbf{v}=(0.2,0.1,-1.8)$ m/s (Vertical descent profile)
Acoustic Array: Local high-frequency rotor harmonics detected (4.8 kHz acoustic spike)
Classification: ROTARY_AERIAL_INCURSION

Secondary Vector (Feline Payload):

Impact deceleration: Sudden kinetic contact at ground plane ($z=0$)
Aspect Ratio: Rapid shift from compact bundle to quadruped stalking gait
Bounding Box Footprint: 0.18 m^2
(Feline mass class)
Trajectory: Vector heading directly converges toward hamster swarm centroid ($\theta=42^\circ$)
 $\frac{dx}{dt}$
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Frame 5: The Cost Matrix Execution & Output

Math Overlay (Asymmetric Edge Escalation):

$$\log\left(\frac{\Pr(\text{Hostile Breach} \mid X)}{\Pr(\text{Defensive Counter-Asset} \mid X)}\right)$$

)> τ

override

Edge Action:

Local NPU concludes: **Aerial drop of biological counter-measure targeting Class 0/Micro-Fauna.**

High-voltage perimeter alarms suppressed; physical perimeter bollards remain locked.

Telemetry Payload: Single 64-byte compact Protobuf packet sent over 900 MHz mesh:

Protocol Buffers

4

8

12.4

2.1 1.4

64

6.2

Why this kills in a Pitch Deck / Demo Reel:

Multi-Domain Tracking: Proves your telemetry engine handles 3D aerial ingress, vertical drop vectors, and ground target convergence at the same time.

Bandwidth Punchline: The cloud system uploaded half a gigabyte of 4K video to tell security that "something moved." VectorEdge parsed an entire tactical air-drop and ground pursuit on a \$15 edge chip, used **64 bytes of bandwidth**, and took **6 milliseconds**.